

# Ashutosh Modi

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## EDUCATION

### University of North Carolina at Charlotte

*B.S. in Data Science*

- Honors: Dean's List; PILOT Program; Association for Computing Machinery

Charlotte, NC

Aug. 2025 – May 2027

### Wake Technical Community College

*Associate's Degree in Science*

- SASA Club – Membership Coordinator

Cary, NC

Aug. 2023 – May 2025

## TECHNICAL SKILLS

**Languages:** Python, SQL, R, Java, C, TypeScript

**Data Engineering:** Apache Spark (PySpark, Structured Streaming), ETL/ELT Pipeline Design, Data Modeling, Window Functions

**ML & Analytics:** pandas, NumPy, GeoPandas, Matplotlib, Power BI, Excel, LLM Integration (Ollama, Groq)

**Databases & Backend:** PostgreSQL, MySQL, SQLite, SQLAlchemy, FastAPI, REST API Design, JWT/OAuth

**Platforms & Tools:** Git, GitHub, Next.js, Streamlit, Tailwind CSS, Power Platform (Power Apps, SharePoint), n8n

## EXPERIENCE

### Teaching Assistant, DTSC 1301: Data and Society

Aug 2026 – Present

*University of North Carolina at Charlotte*

Charlotte, NC

- Support a 106+ student studio course on the Python data science workflow, guiding students through Pandas, GeoPandas, exploratory analysis, and visualization alongside the ethics of data use.
- Grade 800+ weekly quizzes, assignments, and in-class activities, applying consistent rubrics and delivering feedback on data cleaning, statistical interpretation, and visualization design.

### Teaching Assistant, SVSM Program

June 2026 – Aug 2026

*University of North Carolina at Charlotte*

Charlotte, NC

- Supported a competitive summer STEM program for academically advanced rising high school juniors and seniors selected statewide for academic ability and motivation.
- Guided 50+ students through real-world data analysis topics – civic data, public safety, housing equity, EV infrastructure – using Charlotte open datasets to build practical data literacy.

### Data Steward

June 2026 – Aug 2026

*University of North Carolina at Charlotte*

Charlotte, NC

- Partnered with RWCI, a nonprofit, to design a client progress tracking system using Power Apps, SharePoint Lists, and Power BI, replacing manual tracking with a structured data pipeline.
- Built a Power BI dashboard applying the FNI (Family Needs Index) framework, translating meeting notes into numeric progress scores, and delivered trend insights – average scores, high-support flags – to program leadership.

## PROJECTS

### CareerCore | *Next.js, FastAPI, Python, SQLite, Groq AI*

Apr. 2026

- Built a full-stack career intelligence platform serving resume analysis, skill-gap tracking, and interview practice through a FastAPI backend with JWT auth and LinkedIn OAuth.
- Engineered a resume scoring algorithm scoring uploaded PDFs against job descriptions with a 55/25/20 weighting across required skills, preferred skills, and quantified impact.
- Integrated Groq LLaMA 3.3 70B with structured JSON output to generate bullet rewrites, JD summaries, and automated interview question grading.

### VitalStream | *Python, Apache Kafka, Spark, PostgreSQL, FastAPI, Ollama*

June 2026

- Built a real-time ICU monitoring system streaming vitals through Kafka into a Spark Structured Streaming pipeline, scoring patient risk every 5 minutes via the clinically validated NEWS2 framework.
- Engineered a NEWS2 scoring engine in PySpark evaluating five physiological parameters to classify patients as LOW, MEDIUM, or HIGH risk in real time.
- Integrated a local LLM (Ollama) to generate real-time clinical reasoning for high-risk patients and validated the pipeline against de-identified MIMIC-IV ICU data.

### Smart Car Insurance | *Python, pandas, FastAPI, PostgreSQL, Next.js, n8n*

July 2026

- Built a usage-based auto insurance platform transforming telematics and policy data into gold analytics tables, serving driver risk scores and AI-assisted claims via FastAPI + Next.js.
- Engineered a telematics feature pipeline that sessionizes trips and computes daily driver risk scores with LOW/MEDIUM/HIGH risk bands for portfolio analytics.
- Implemented an AI claims workflow routing accident photos and police reports through an n8n webhook for damage/fault/cost estimation with human-in-the-loop adjuster review.